

Electrodynamics

Problem Set PS.18

Wavefunctions
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1. Time Evolution in Infinite Square Well

Consider a particle of mass m in a one-dimensional infinite square well extending from $x = 0$ to $x = L$ with potential:

$$v(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{otherwise} \end{cases}$$

We know the normalized stationary states and energy eigenvalues are given by:

$$\phi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right)$$

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}$$

at time $t = 0$ the particle is prepared in the normalized state:

$$\psi(x) = \begin{cases} Ax & 0 \leq x \leq L/2 \\ A(L-x) & L/2 \leq x \leq L \\ 0 & \text{elsewhere} \end{cases}$$

a. Determine the normalization constant A .

$$\int_0^L |\psi(x)|^2 dx = 1$$
$$\int_0^{L/2} (Ax)^2 dx + \int_{L/2}^L (A(L-x))^2 dx = 1$$

By symmetry:

$$2A^2 \int_0^{L/2} x^2 dx = 1$$
$$2A^2 \left[\frac{x^3}{3} \right]_0^{L/2} = 2A^2 \left(\frac{L^3/8}{3} \right) = A^2 \frac{L^3}{12} = 1$$
$$\therefore \boxed{A = \sqrt{\frac{12}{L^3}} = \frac{2\sqrt{3}}{L^{3/2}}}$$

b. We know that a general state can be written as the linear combination of eigenstates:

$$\psi(x) = \sum_{n=1}^{\infty} c_n \phi_n(x)$$

Where

$$c_n = \int_0^L \phi_n^*(x) \psi(x) dx$$

Explicitly calculate c_1, c_2, c_3 .

$$c_n = \sqrt{\frac{2}{L}} A \left[\int_0^{L/2} x \sin\left(\frac{n\pi x}{L}\right) dx + \int_{L/2}^L (L-x) \sin\left(\frac{n\pi x}{L}\right) dx \right]$$

$$I_1 = \left[-\frac{Lx}{n\pi} \cos\left(\frac{n\pi x}{L}\right) + \frac{L^2}{n^2\pi^2} \sin\left(\frac{n\pi x}{L}\right) \right]_0^{L/2}$$

$$I_1 = -\frac{L^2}{2n\pi} \cos\left(\frac{n\pi}{2}\right) + \frac{L^2}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)$$

$$\int_{L/2}^0 u \sin\left(\frac{n\pi(L-u)}{L}\right) (-du) = \int_0^{L/2} u \sin\left(n\pi - \frac{n\pi u}{L}\right) du$$

- If n is **even**: $1 + (-1)^{n+1} = 0$, so $c_{\text{even}} = 0$.
- If n is **odd**: $1 + (-1)^{n+1} = 2$. Also, for odd n , $\cos(n\pi/2) = 0$.

When n is odd:

$$c_n = \sqrt{\frac{2}{L}} \sqrt{\frac{12}{L^3}} \times 2 \left(\frac{L^2}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) \right) = \frac{\sqrt{24}}{L^2} \frac{2L^2}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right) = \frac{4\sqrt{6}}{n^2\pi^2} \sin\left(\frac{n\pi}{2}\right)$$

$$c_1 = \frac{4\sqrt{6}}{1^2\pi^2} \sin\left(\frac{\pi}{2}\right) = \frac{4\sqrt{6}}{\pi^2}$$

$$c_2 = 0$$

$$c_3 = \frac{4\sqrt{6}}{3^2\pi^2} \sin\left(\frac{3\pi}{2}\right) = -\frac{4\sqrt{6}}{9\pi^2}$$

c. Write out the time-dependent state $\psi(x, t)$ in the form:

$$\psi(x, t) = \sum_{n=1}^3 c_n \phi_n(x) e^{-iE_n t/\hbar}$$

$$\psi(x, t) = \left(\frac{4\sqrt{6}}{\pi^2} \right) \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right) e^{-iE_1 t/\hbar} - \left(\frac{4\sqrt{6}}{9\pi^2} \right) \sqrt{\frac{2}{L}} \sin\left(\frac{3\pi x}{L}\right) e^{-iE_3 t/\hbar}$$

$$= \frac{8\sqrt{3}}{\pi^2\sqrt{L}} \sin\left(\frac{\pi x}{L}\right) e^{-iE_1 t/\hbar} - \frac{8\sqrt{3}}{9\pi^2\sqrt{L}} \sin\left(\frac{3\pi x}{L}\right) e^{-iE_3 t/\hbar}$$

d. Suppose an energy measurement is made at time $t = 0$.

i. What are the possible results?

The energy eigenvalues $E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}$. However, since $c_n = 0$ for all even integers, it's really only E_n for odd n .

ii. What is the probability of obtaining the result E_n ?

$$P(E_n) = |c_n|^2 = \begin{cases} \frac{96}{n^4\pi^4} & \text{for } n \text{ odd} \\ 0 & \text{for } n \text{ even} \end{cases}$$

iii. What is the probability of measuring E_1, E_2, E_3 ?

$$P(E_1) = |c_1|^2 = \frac{96}{\pi^4} \approx 0.9855 \text{ (or 98.55\%)}$$

$$P(E_2) = |c_2|^2 = 0$$

$$P(E_3) = |c_3|^2 = \frac{96}{81\pi^4} = \frac{32}{27\pi^4} \approx 0.0122 \text{ (or 1.22\%)}$$

e. Suppose the first energy measurement gives the value E_3 .

i. What is the state immediately after the measurement?

$$\psi_{\text{after}}(x) = \phi_3(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{3\pi x}{L}\right)$$

ii. What is the state at a later time $t + \Delta t$?

$$\psi_{\text{later}}(x, t + \Delta t) = \phi_3(x) e^{-iE_3\Delta t/\hbar} = \sqrt{\frac{2}{L}} \sin\left(\frac{3\pi x}{L}\right) e^{-iE_3\Delta t/\hbar}$$

f. Calculate $\langle H \rangle$. Should this be time-dependent?

$$\langle H \rangle = \sum_{n \text{ odd}} \left(\frac{96}{n^4\pi^4} \right) \left(\frac{n^2\pi^2\hbar^2}{2mL^2} \right) = \frac{48\hbar^2}{\pi^2mL^2} \sum_{n \text{ odd}} \frac{1}{n^2}$$

$$\langle H \rangle = \frac{48\hbar^2}{\pi^2mL^2} \left(\frac{\pi^2}{8} \right) = \frac{6\hbar^2}{mL^2}$$

No. The potential is time-independent, which implies energy is conserved.

g. Calculate $\langle x \rangle$. Should this be time-dependent?

At $t = 0$, the probability distribution $|\psi(x, 0)|^2$ is perfectly symmetric about the center of the well, so $\langle x \rangle = L/2$.

Therefore, $\langle x \rangle = L/2$ permanently. **No, it is not time-dependent.**

h. What if we make a position measurement of the state $\psi(x, t)$ (not an energy measurement)? Explain the state of the system just after this measurement. Explicitly comment on its energy content and if the state is stationary or non-stationary.

State just after measurement: Measuring the position collapses the wavefunction into a Dirac delta function spiked exactly at the measured position x_0 , so $\psi_{\text{after}}(x) = \delta(x - x_0)$.

Energy content: Heisenberg's basics tell us that knowing the position causes infinite uncertainty in momentum, which corresponds to an infinite amount of kinetic energy. $\langle H \rangle \rightarrow \infty$.

Stationary or non-stationary? It is **non-stationary**. The delta function is a superposition of nearly all possible energy eigenstates. As time passes, the phase factors of different modes will rapidly interfere and reduce, leaving the wave function to take over the well.